When is “if” true?

Horos · a reading and experiment by Anthony Mercer.

> A finite compatibility model: p and not-q conflict exactly when no admitted case contains both. The admitted cases are supplied assumptions, not observations that establish necessity. This does not reconstruct ancient connection, relevance, Diodorus's temporal/modal criterion, or potential containment.

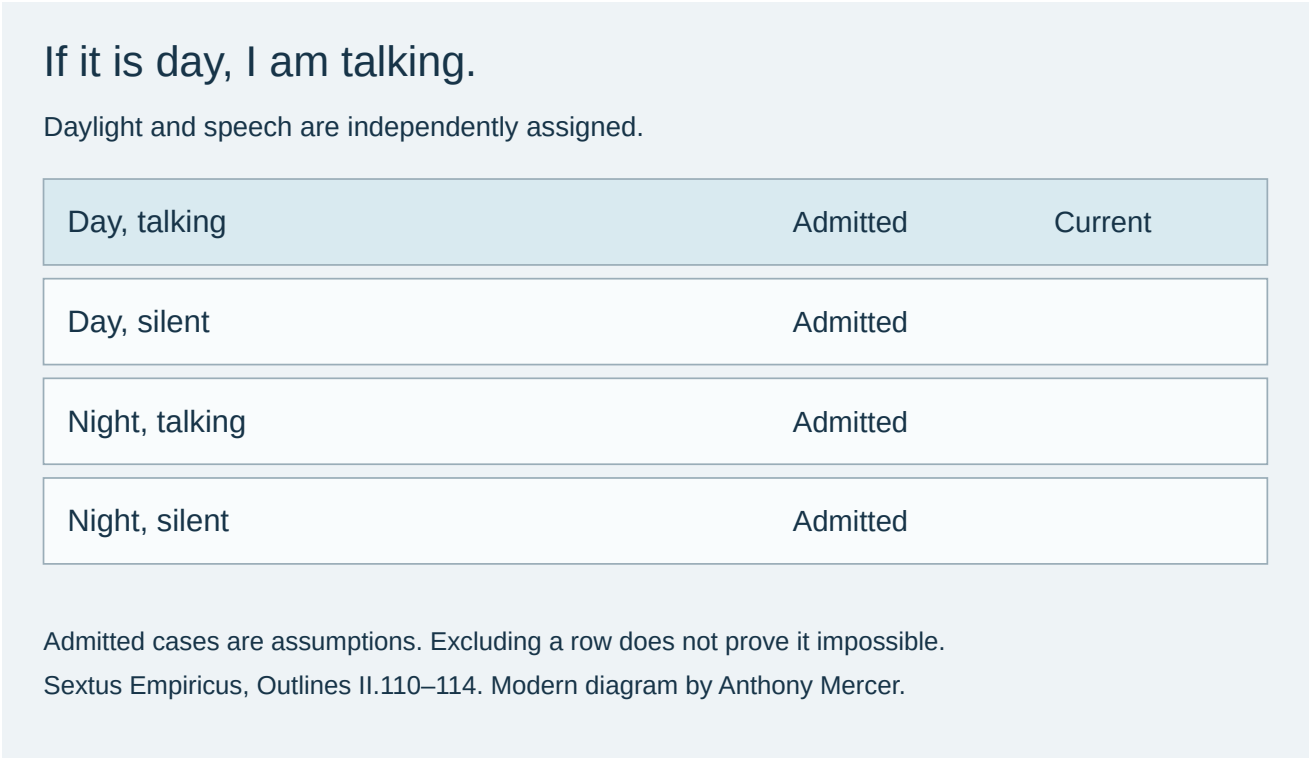
Your explanation

Both current clauses are true, so Philo’s criterion accepts this pair. The pinned table also admits day with silence, which defeats the finite compatibility test. Removing that case makes my new table pass but adds an assumption I have not justified.
The Greek μάχηται asks for a conflict. A checkbox alone does not establish it.

The snapshots use the same sentence. Compare changes to the current facts and to the admitted cases separately.

Pinned experiment

If it is day, I am talking.



Current facts and admitted cases

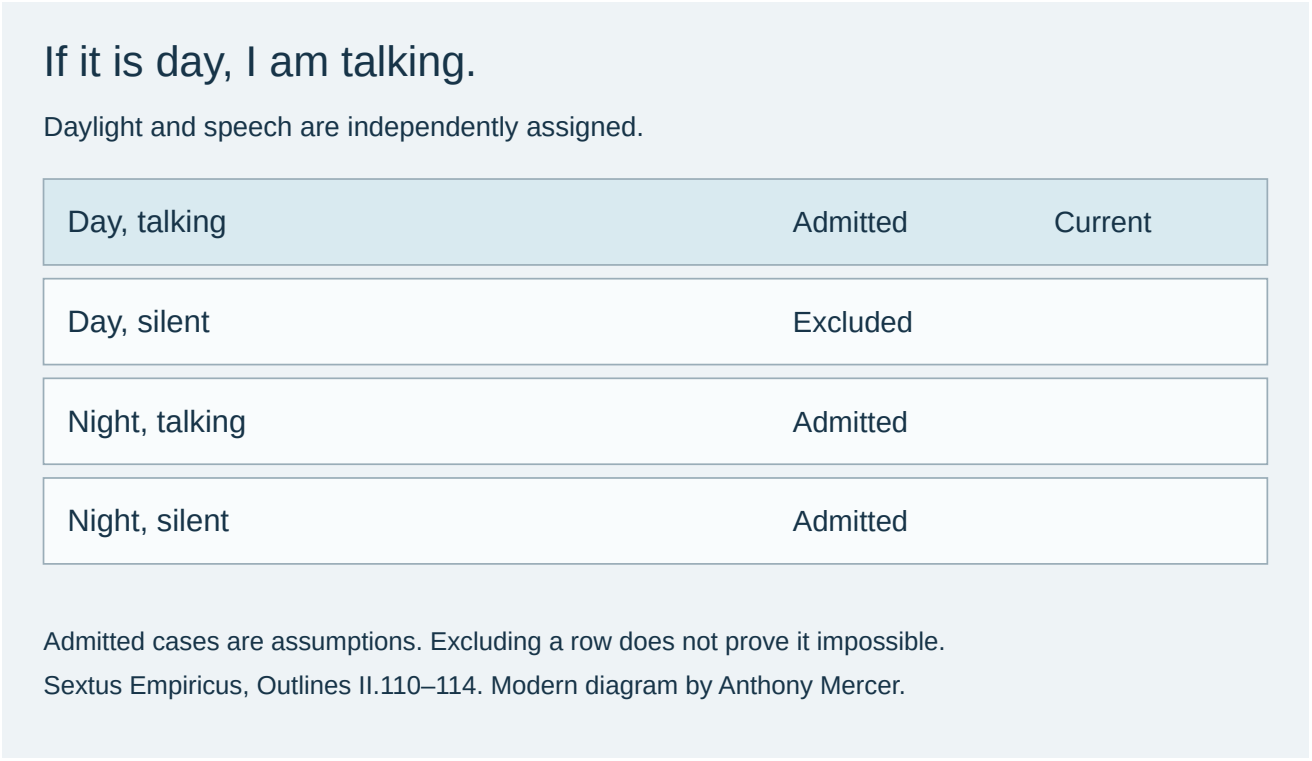
Case	Antecedent	Consequent	Admitted?	Current?
Day, talking	True	True	Yes	Yes
Day, silent	True	False	Yes	
Night, talking	False	True	Yes	
Night, silent	False	False	Yes	

Philo, at the current facts: true. The current pair is not true-antecedent / false-consequent.

Finite compatibility: fails. Witness: Day, silent.

Current experiment

If it is day, I am talking.



Current facts and admitted cases

Case	Antecedent	Consequent	Admitted?	Current?
Day, talking	True	True	Yes	Yes
Day, silent	True	False	No	
Night, talking	False	True	Yes	
Night, silent	False	False	Yes	

Philo, at the current facts: true. The current pair is not true-antecedent / false-consequent.

Finite compatibility: passes. No admitted case contains a true antecedent and false consequent.

What the two tests establish

Philo's reported criterion is truth-functional: only a true antecedent with a false consequent makes the conditional false. It requires no causal connection. The finite compatibility test asks whether that pair is admitted anywhere in the supplied table. These are different questions.

The four combinations exhaust two independently bivalent clauses, not the world's possibilities. Repeating ‘it is day’ gives the same statement twice, so its truth values must agree. Removing ‘day, silent’ by hand adds an assumption; it is not evidence that the combination cannot occur. An empty table and a table with no antecedent-true cases are explicitly diagnosed.

Sextus reports four accounts. This experiment computes no verdict for Diodorus or the potential-containment account. It does not choose a winning school. The report itself challenges how a disputed consequence relation could certify its own criterion.

Philo: one forbidden combination

Outlines II.110 · complete printed page 92

ὁ μὲν γὰρ Φίλων φησὶν ὑγιὲς εἶναι συνημμένον τὸ μὴ ἀρχόμενον ἀπὸ ἀληθοῦς καὶ λῆγον ἐπὶ ψεῦδος, οἷον ἡμέρας οὔσης καὶ ἐμοῦ διαλεγομένου τὸ ‘εἰ ἡμέρα ἔστιν, ἐγὼ διαλέγομαι’.

Close English

Philo says that a sound conditional is one that does not begin from something true and end in something false: for example, when it is day and I am talking, ‘If it is day, I am talking.’

Teaching Latin · new translation

Philo sanum esse conexum dicit quod non a vero incipit et in falsum desinit; velut, cum dies sit et ego disseram: ‘Si dies est, ego dissero.’

‘Begins’ and ‘ends’ name the clauses, not events occurring in sequence. Here ὑγιές means sound or acceptable as a conditional; this is different from the validity of an argument with several premises. Talking translates διαλέγομαι, which can mean conversing or discussing.

Contextual phrase notes

Greek	Teaching Latin	Meaning here
συνημμένον	conexum	synēmménon: something joined together; here a conditional statement. A perfect passive participle used as a noun.
ὑγιὲς	sanum	hygiès: sound, literally healthy. Sextus is reporting rival standards for an acceptable conditional.
ἀπὸ ἀληθοῦς	a vero	apò alēthoûs: from something true. The first clause is the antecedent; it is not an earlier event.
ἐπὶ ψεῦδος	in falsum	epì pseûdos: to something false. Philo excludes the true-antecedent / false-consequent combination.
ἡμέρας οὔσης	cum dies sit	hēméras oúsēs: ‘it being day’. This genitive absolute sets the facts for the example.
διαλέγομαι	dissero	dialégomai: I converse or discuss. The workshop shortens this to ‘I am talking’. It does not mean that daylight causes the speech.

Diodorus: more than this moment

Outlines II.110–111, excerpt · complete printed page 92

ὁ δὲ Διόδωρος, ὃ μήτε ἐνεδέχετο μήτε ἐνδέχεται ἀρχόμενον ἀπὸ ἀληθοῦς λήγειν ἐπὶ ψεῦδος· καθ’ ὃν τὸ μὲν εἰρημένον συνημμένον ψεῦδος εἶναι δοκεῖ, ἐπεὶ ἡμέρας μὲν οὔσης ἐμοῦ δὲ σιωπήσαντος ἀπὸ ἀληθοῦς ἀρξάμενον ἐπὶ ψεῦδος καταλήξει.

Close English

Diodorus, however, [calls sound] one which neither was nor is capable of beginning from something true and ending in something false. On his account the conditional just mentioned seems false: when it is day but I have fallen silent, it will begin from something true and end in something false.

Teaching Latin · new translation

Diodorus autem id [sanum dicit] quod neque poterat neque potest a vero incipiens in falsum desinere. Secundum eum illud conexum falsum esse videtur: die enim manente, si tacuerim, a vero incipiens in falsum desinet.

The words in square brackets complete the elliptical Greek in English and Latin. Sextus's criterion includes past and present possibility, and his example invokes becoming silent. A finite table of chosen cases cannot certify this temporal and modal claim. The workshop therefore gives Diodorus no computed verdict.

Contextual phrase notes

Greek	Teaching Latin	Meaning here
μήτε ἐνεδέχετο μήτε ἐνδέχεται	neque poterat neque potest	mḗte enedékheto mḗte endékhetai: neither was it possible nor is it possible. The tense change is part of the criterion, not an instruction to count a few observed moments.
σιωπήσαντος	si tacuerim	siōpḗsantos: having fallen silent. Day may remain while speech stops; the first clause can stay true as the second becomes false.

A deliberately uncomfortable example

Outlines II.111, excerpt · complete printed page 92

ἐκεῖνο δὲ ἀληθές ‘εἰ οὐκ ἔστιν ἀμερῆ τῶν ὄντων στοιχεῖα, ἔστιν ἀμερῆ τῶν ὄντων στοιχεῖα’· ἀεὶ γὰρ ἀπὸ ψεύδους ἀρχόμενον τοῦ ‘οὐκ ἔστιν ἀμερῆ τῶν ὄντων στοιχεῖα’ εἰς ἀληθὲς καταλήξει κατ’ αὐτὸν τὸ ‘ἔστιν ἀμερῆ τῶν ὄντων στοιχεῖα’.

Close English

But this [conditional seems] true: ‘If there are no partless elements of existing things, there are partless elements of existing things.’ For on his account it will always begin from the false ‘There are no partless elements of existing things’ and end in the true ‘There are partless elements of existing things.’

Teaching Latin · new abridgment

Illud autem [conexum verum videtur]: ‘Si non sunt rerum elementa sine partibus, sunt rerum elementa sine partibus.’ Secundum eum enim semper a falso incipiet et in verum desinet.

Sextus reports what is true ‘on his account’ (Diodorus's); the workshop does not assert that ancient atomism or its negation is true. This example separates a fixed false antecedent from the alleged connection. It is preserved in the reading but is not assigned an empirical truth value by the controls. The Latin condenses the repeated quoted clauses.

Contextual phrase notes

Greek	Teaching Latin	Meaning here
ἀμερῆ	sine partibus	amerê: without parts, partless. This reports the example's philosophical vocabulary, rather than measurements of modern physical atoms.
κατ’ αὐτὸν	secundum eum	kat’ autòn: according to him. The attribution prevents a reported premise from becoming the reader's own assertion.

Connection: try the opposite of the ending

Outlines II.111–112, excerpt · complete printed page 92

οἱ δὲ τὴν συνάρτησιν εἰσάγοντες ὑγιὲς εἶναί φασι συνημμένον, ὅταν τὸ ἀντικείμενον τῷ ἐν αὐτῷ λήγοντι μάχηται τῷ ἐν αὐτῷ ἡγουμένῳ· καθ’ οὓς τὰ μὲν εἰρημένα συνημμένα ἔσται μοχθηρά, ἐκεῖνο δὲ ἀληθές ‘εἰ ἡμέρα ἔστιν, ἡμέρα ἔστιν’.

Close English

Those who introduce connection say that a conditional is sound when the opposite of its consequent conflicts with its antecedent. On their account the conditionals mentioned above will be bad, but this one true: ‘If it is day, it is day.’

Teaching Latin · new translation

Qui autem conexionem introducunt sanum esse conexum dicunt, cum id quod consequenti opponitur cum antecedente pugnet. Secundum eos superiora conexa vitiosa erunt, hoc autem verum: ‘Si dies est, dies est.’

Sextus does not name these proponents here. The finite experiment represents ‘conflicts’ by excluding every admitted case with a true antecedent and false consequent. That is an explicit modern compatibility model, not a complete reconstruction of ancient connection, relevance, or necessity. Choosing the table is an additional assumption.

Contextual phrase notes

Greek	Teaching Latin	Meaning here
συνάρτησιν	conexionem	synártēsin: connection or coherence. The question is stronger than whether the two clauses happen to agree right now.
τὸ ἀντικείμενον	id quod opponitur	tò antikeímenon: what is opposed. In the experiment the opposite of ‘I am talking’ is ‘I am not talking’.
λήγοντι	consequenti	lḗgonti: the ending clause, the consequent. The dative goes with ‘what is opposed to’.
μάχηται	pugnet	mákhētai: conflicts or fights. The model tests co-admission in a finite case set; it cannot independently decide what is truly incompatible.
ἡγουμένῳ	antecedente	hēgouménōi: the leading clause, the antecedent. A true antecedent with a false consequent is the witness to look for.

A fourth account resists the easy answer

Outlines II.112–113, excerpt · complete printed page 92

οἱ δὲ τῇ ἐμφάσει κρίνοντές φασιν ὅτι ἀληθές ἐστι συνημμένον οὗ τὸ λῆγον ἐν τῷ ἡγουμένῳ περιέχεται δυνάμει· καθ’ οὓς τὸ ‘εἰ ἡμέρα ἔστιν, ἡμέρα ἔστι’ καὶ πᾶν τὸ διαφορούμενον [ἀξίωμα] συνημμένον ἴσως ψεῦδος ἔσται· αὐτὸ γάρ τι ἐν ἑαυτῷ περιέχεσθαι ἀμήχανον.

Close English

Those who judge by indication say that a conditional is true when its consequent is contained potentially in its antecedent. On their account, ‘If it is day, it is day’ and every repeated conditional will perhaps be false: for a thing's being contained in itself is impossible.

Teaching Latin · new translation

Qui ex indicatione iudicant verum esse conexum dicunt cuius consequens in antecedente potentia continetur. Secundum eos ‘Si dies est, dies est’ et omne conexum repetitum fortasse falsum erit: nihil enim in se ipso contineri potest.

‘Indication’ translates the difficult ἔμφασις here; ‘force’ or ‘meaning’ are other renderings. Sextus presents this as another reported account, and uses ‘perhaps’ for his objection. His report of its treatment of the repeated conditional must not be silently replaced by the modern identity result. This room computes no verdict for this account. The editor brackets ἀξίωμα for deletion; the English omits that bracketed noun. ‘Repeated’ is an interpretive rendering of διαφορούμενον in light of the day/day example, not its unambiguous everyday meaning. The apparatus records διφορούμενον in EAB and a different spelling in L; the displayed Greek follows the printed main text.

Contextual phrase notes

Greek	Teaching Latin	Meaning here
διαφορούμενον	repetitum	diaphoroúmenon: rendered ‘repeated’ here from the day/day example. The main text prints διαφορούμενον; the apparatus at printed page 92, line 17 records διφορούμενον in EAB and διαφορόμενον in L (with an erased letter). Read this as a textual and interpretive difficulty, not a settled vocabulary equivalence.
ἐμφάσει	indicatione	emphásei: by indication, force, or meaning. Translating the term does not settle which precise relation its proponents intended.
δυνάμει	potentia	dynámei: potentially, in capacity. This is not a numerical ‘power’ used by the program.
ἴσως	fortasse	ísōs: perhaps. Keep the qualification in Sextus's reported objection.

Who gets to choose the test?

Outlines II.114, excerpt · complete printed page 93

ζητουμένου δὲ περὶ τοῦ πῶς κρινοῦμεν τὴν ἀκολουθίαν τοῦ λήγοντος πρὸς τὸ ἡγούμενον, ὁ διάλληλος εὑρίσκεται τρόπος. ἵνα μὲν γὰρ ἡ κρίσις τοῦ συνημμένου ἀποδειχθῇ, τὸ συμπέρασμα τοῖς λήμμασι τῆς ἀποδείξεως ἀκολουθεῖ, ὡς προειρήκαμεν· ἵνα δὲ πάλιν τοῦτο πιστευθῇ, δεῖ τὸ συνημμένον καὶ τὴν ἀκολουθίαν ἐπικεκρίσθαι· ὅπερ ἄτοπον.

Close English

When it is being asked how we shall judge the consequent's following from the antecedent, the circular mode is found. To prove the criterion for the conditional, the conclusion follows from the proof's premises, as we said before; but for that in turn to be trusted, the conditional and the following must already have been judged. That is absurd.

Teaching Latin · new abridgment

Cum quaeratur quomodo consequentiam iudicemus, modus circularis invenitur. Ut enim iudicium conexi probetur, conclusio probationis ex praemissis sequitur; ut autem hoc credatur, conexum et consequentia iam iudicata esse debent. Quod absurdum est.

This is Sextus's challenge to the use of a disputed consequence relation to certify its own criterion. It is not the workshop declaring that all logic or proof is impossible. The Latin is an abridgment. The program's finite search proves a result only after its modern rules and considered cases have been fixed.

Contextual phrase notes

Greek	Teaching Latin	Meaning here
ἀκολουθίαν	consequentiam	akolouthían: following or consequence. The dispute asks what entitles us to say the consequent follows.
διάλληλος	circularis	diállēlos: reciprocal or circular. Here a proof appeals to the very relation whose test is disputed.

Edition and text policy

Sextus Empiricus, Outlines of Pyrrhonism (Πυρρώνειοι ὑποτυπώσεις) II.110–114. Hermann Mutschmann, Sexti Empirici Opera, volume I (Leipzig: B. G. Teubner, 1912), complete printed pages 92–93. This is a modern printed witness to an ancient report, not a surviving work by Philo or Diodorus.

The Greek was checked against the retained printed pages. Line endings are joined; punctuation and typographic apostrophes are normalized. Square brackets in the fourth account retain the editor's deletion of ἀξίωμα. The English and teaching Latin are new. The Latin is a teaching translation, not an ancient or Renaissance witness. No independent scholarly review is claimed.

Mutschmann 1912 · complete printed page 92

81

ὅτε λέγω τὸ ‘φῶς ἔστιν’, οὐκέτι ἔστι τὸ ‘ἡμέρα ἔστιν’. εἰ οὖν τὰ μὲν συγκείμενα ἔκ τινων ἀδύνατον ὑπάρχειν μὴ συνυπαρχόντων ἀλλήλοις τῶν μερῶν αὐτῶν, τὰ δὲ ἐξ ὧν σύγκειται τὸ ἀξίωμα οὐ συνυπάρχει ἀλλήλοις, οὐχ ὑπάρξει
110 τὸ ἀξίωμα. ἵνα δὲ καὶ ταῦτα παραλίπωμεν, τὸ ὑγιὲς συνημμένον ἀκατάληπτον εὑρεθήσεται. ὁ μὲν γὰρ Φίλων φησὶν ὑγιὲς εἶναι συνημμένον τὸ μὴ ἀρχόμενον ἀπὸ ἀληθοῦς καὶ λῆγον ἐπὶ ψεῦδος, οἷον ἡμέρας οὔσης καὶ ἐμοῦ διαλεγομένου τὸ ‘εἰ ἡμέρα ἔστιν, ἐγὼ διαλέγομαι’, ὁ δὲ Διόδωρος, ὃ μήτε ἐνεδέχετο μήτε ἐνδέχεται ἀρχόμενον ἀπὸ ἀληθοῦς λήγειν ἐπὶ ψεῦδος· καθ’ ὃν τὸ μὲν εἰρημένον συνημμένον ψεῦδος εἶναι δοκεῖ, ἐπεὶ ἡμέρας μὲν οὔσης ἐμοῦ δὲ σιωπήσαντος ἀπὸ ἀληθοῦς ἀρξάμενον ἐπὶ ψεῦδος καταλήξει, ἐκεῖνο δὲ ἀληθές ‘εἰ οὐκ
111 ἔστιν ἀμερῆ τῶν ὄντων στοιχεῖα, ἔστιν ἀμερῆ τῶν ὄντων στοιχεῖα’· ἀεὶ γὰρ ἀπὸ ψεύδους ἀρχόμενον τοῦ ‘οὐκ ἔστιν ἀμερῆ τῶν ὄντων στοιχεῖα’ εἰς ἀληθὲς καταλήξει κατ’ αὐτὸν τὸ ‘ἔστιν ἀμερῆ τῶν ὄντων στοιχεῖα’. οἱ δὲ τὴν συνάρτησιν εἰσάγοντες ὑγιὲς εἶναί φασι συνημμένον, ὅταν τὸ ἀντικείμενον τῷ ἐν αὐτῷ λήγοντι μάχηται τῷ ἐν αὐτῷ ἡγουμένῳ· καθ’ οὓς τὰ μὲν εἰρημένα συνημμένα ἔσται μοχθηρά, ἐκεῖνο δὲ ἀληθές ‘εἰ ἡμέρα ἔστιν, ἡμέρα
112 ἔστιν’. οἱ δὲ τῇ ἐμφάσει κρίνοντές φασιν ὅτι ἀληθές ἐστι συνημμένον οὗ τὸ λῆγον ἐν τῷ ἡγουμένῳ περιέχεται δυνάμει· καθ’ οὓς τὸ ‘εἰ ἡμέρα ἔστιν, ἡμέρα ἔστι’ καὶ πᾶν τὸ διαφορούμενον [ἀξίωμα] συνημμένον ἴσως ψεῦδος ἔσται·
113 αὐτὸ γάρ τι ἐν ἑαυτῷ περιέχεσθαι ἀμήχανον. ταύτην τοίνυν τὴν διαφωνίαν ἐπικριθῆναι ἀμήχανον ἴσως εἶναι δόξει. οὔτε γὰρ ἀναποδείκτως προκρίνοντές τινα τῶν στάσεων τῶν προειρημένων πιστοὶ ἐσόμεθα οὔτε μετὰ ἀποδείξεως. καὶ γὰρ ἡ ἀπόδειξις ὑγιὴς εἶναι δοκεῖ, ὅταν ἀκολουθῇ τῇ διὰ

§ 110 ~ adv. dogm. II 113—117.

28 *οὐ συνυπάρξει* EAB **31** *μὴ* secl. Kayser **10** *συνάγοντες* L **12** *predicta* T **17** *τὸ* om. edd. ‖ *διαφορούμενον* MEAB: *διφορόμενον* (eras. *α*) L: *omne quod diversificatur* T ‖ *ἀξίωμα* del. Bekk. **19** post *ἴσως* add. *ἂν* L **20** *opinionum* T (*φάσεων*, quod iam coni. Gen.)

Public-domain printed edition. Original retained page rendition (https://archive.org/download/sextiempiriciope01sext/page/n123_w1800.jpg).

Mutschmann 1912 · complete printed page 93

82 τῶν λημμάτων αὐτῆς συμπλοκῇ τὸ συμπέρασμα αὐτῆς ὡς
λῆγον ἡγουμένῳ, οἷον οὕτως εἰ ἡμέρα ἔστιν, φῶς ἔστιν·
25 ἀλλὰ μὴν ἡμέρα ἔστιν· φῶς ἄρα ἔστιν. εἴπερ ἡμέρα ἔστι,
φῶς ἔστιν· καὶ ἡμέρα ἔστι καὶ φῶς ἔστιν'. ζητουμένου δὲ 114
περὶ τοῦ πῶς κρινοῦμεν τὴν ἀκολουθίαν τοῦ λήγοντος
πρὸς τὸ ἡγούμενον, ὁ διάλληλος εὑρίσκεται τρόπος. ἵνα
μὲν γὰρ ἡ κρίσις τοῦ συνημμένου ἀποδειχθῇ, τὸ συμ-
30 πέρασμα τοῖς λήμμασι τῆς ἀποδείξεως ἀκολουθεῖ, ὡς προει-
ρήκαμεν· ἵνα δὲ πάλιν τοῦτο πιστευθῇ, δεῖ τὸ συνημμέ-
83 νον καὶ τὴν ἀκολουθίαν ἐπικεκρίσθαι· ὅπερ ἄτοπον. ἀκα- 115
τάληπτον ἄρα τὸ ὑγιὲς συνημμένον. ἀλλὰ καὶ τὸ
προκαθηγούμενον ἄπορόν ἐστιν. τὸ μὲν γὰρ προκα-
θηγούμενον, ὥς φασιν, ἐστὶ τὸ ἡγούμενον ἐν τοι-
5 ούτῳ συνημμένῳ, ὃ ἄρχεται ἀπὸ ἀληθοῦς καὶ
λήγει ἐπὶ ἀληθές. εἰ δὲ ἐκκαλυπτικόν ἐστι τοῦ λήγοντος 116
τὸ σημεῖον, ἤτοι πρόδηλόν ἐστι τὸ λῆγον ἢ ἄδηλον. εἰ μὲν
οὖν πρόδηλον, οὐδὲ τοῦ ἐκκαλύψοντος δεήσεται, ἀλλὰ συγ-
καταληφθήσεται αὐτῷ, καὶ οὐκ ἔσται αὐτοῦ σημειωτόν, διόπερ
10 οὐδὲ ἐκεῖνο τούτου σημεῖον. εἰ δὲ ἄδηλον, ἐπεὶ περὶ τῶν ἀδή-
λων διαπεφώνηται ἀνεπικρίτως, ποῖα μέν ἐστιν αὐτῶν ἀληθῆ
ποῖα δὲ ψευδῆ, καὶ ὅλως εἰ ἔστι τι αὐτῶν ἀληθές, ἄδηλον
ἔσται, εἰ εἰς ἀληθὲς λήγει τὸ συνημμένον. ᾧ συνεισέρχεται καὶ
τὸ ἄδηλον εἶναι εἰ προκαθηγεῖται τὸ ἐν αὐτῷ ἡγούμενον.
15 ἵνα δὲ καὶ ταῦτα παραλίπωμεν, οὐ δύναται ἐκκαλυπτικὸν 117
εἶναι τοῦ λήγοντος, εἴγε πρὸς τὸ σημεῖόν ἐστι τὸ σημειω-
τὸν καὶ διὰ τοῦτο συγκαταλαμβάνεται αὐτῷ. τὰ γὰρ πρός
τι ἀλλήλοις συγκαταλαμβάνεται· καὶ ὥσπερ τὸ δεξιὸν πρὸ
τοῦ ἀριστεροῦ ὡς δεξιὸν ἀριστεροῦ καταληφθῆναι οὐ δύνα-
20 ται, οὐδὲ ἀνάπαλιν, καὶ ἐπὶ τῶν ἄλλων τῶν πρός τι πα-
ραπλησίως, οὕτως οὐδὲ τὸ σημεῖον πρὸ τοῦ σημειωτοῦ
ὡς σημειωτοῦ καταληφθῆναι δυνατὸν ἔσται. εἰ δ' οὐ προ- 118
καταλαμβάνεται τὸ σημεῖον τοῦ σημειωτοῦ, οὐδὲ ἐκκαλυπτικὸν

§§ 117—118 (= 119—120) ~ adv. dogm. II 165.

26 καὶ ἡμέρα — ἔστιν om. T **27** περὶ τοῦ om. T (*quesito autem quomodo*) cf. p. 84, 16 **8** ἐκκαλύψαντος L **9** ἔσται T (*erit*) cf. v. 13: ἔστιν G ‖ σημειωτικόν A **22** ὡς σημειωτοῦ ML ex corr.: *signum ante signatum tamquam signati* T: om. EAB, fortasse scrib. ὡς ⟨σημεῖον⟩ σημειωτοῦ coll. v. 19.

Public-domain printed edition. Original retained page rendition (https://archive.org/download/sextiempiriciope01sext/page/n124_w1800.jpg).

Retained-file identities


```
[
  {
    "path": "LICENSE.txt",
    "bytes": 1071,
    "sha256": "62145e70388e9be739b404f904df6434bacaa748c9b5875c146791457fb83b81",
    "mime": "text/plain"
  },
  {
    "path": "fonts/literata-OFL.txt",
    "bytes": 4389,
    "sha256": "8742963604cd89dc81437811a850018fc03b2bfad686d7422c8235967c87614e",
    "mime": "text/plain"
  },
  {
    "path": "fonts/literata.woff2",
    "bytes": 396664,
    "sha256": "99efd2b91637e922059a38d459abeceade161dc868b47318fa3b9c99137293db",
    "mime": "font/woff2"
  },
  {
    "path": "fonts/provenance.json",
    "bytes": 637,
    "sha256": "a33529cd8d64a41671cbfa367a56f0180d399810f3434f8c00b6e0e07e3115a2",
    "mime": "application/json"
  },
  {
    "path": "fonts/sourcesans3-OFL.txt",
    "bytes": 4579,
    "sha256": "09746787287a289323b0ec3cff4d1a4a801331b82b7207c1e186f5d26619a392",
    "mime": "text/plain"
  },
  {
    "path": "fonts/sourcesans3.woff2",
    "bytes": 170188,
    "sha256": "5f16566f7a40d39b339ad26be151fa5a1ab1f0c2574c7a2e619765584a1acbd8",
    "mime": "font/woff2"
  },
  {
    "path": "sources/conditionals/mutschmann-p92.jpg",
    "bytes": 695470,
    "sha256": "c9ac0eac6934f6b537f25d35e320f1e67d0db3e6e9e9fb77e1e14c607e7ddfed",
    "mime": "image/jpeg",
    "printedPage": 92,
    "source": "https://archive.org/download/sextiempiriciope01sext/page/n123_w1800.jpg",
    "edition": "Mutschmann 1912, Sexti Empirici Opera I",
    "width": 2096,
    "height": 3177
  },
  {
    "path": "sources/conditionals/mutschmann-p93.jpg",
```

```
    "bytes": 749017,
    "sha256": "e48fc4b73dea853b8009454ae1a20cc19ffc9d4dd8728b9f9f14dc9e9cc5bbd1",
    "mime": "image/jpeg",
    "printedPage": 93,
    "source": "https://archive.org/download/sextiempiriciope01sext/page/n124_w1800.jpg",
    "edition": "Mutschmann 1912, Sexti Empirici Opera I",
    "width": 2096,
    "height": 3177
  },
  {
    "path": "sources/conditionals/NOTICE.txt",
    "bytes": 2489,
    "sha256": "50ab6044f839960ec08fa5b615b5059d57a34451794f0f8656c813d73f3085bd",
    "mime": "text/plain"
  }
]
```


Licenses and provenance

LICENSE.txt


```
MIT License

Copyright (c) 2026 Anthony Mercer

Permission is hereby granted, free of charge, to any person obtaining a copy
of this software and associated documentation files (the "Software"), to deal
in the Software without restriction, including without limitation the rights
to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
copies of the Software, and to permit persons to whom the Software is
furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in
all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN
THE SOFTWARE.
```


fonts/literata-OFL.txt


```
Copyright 2017 The Literata Project Authors (https://github.com/googlefonts/literata)

This Font Software is licensed under the SIL Open Font License, Version 1.1.
This license is copied below, and is also available with a FAQ at:
http://scripts.sil.org/OFL
```

SIL OPEN FONT LICENSE Version 1.1 - 26 February 2007

PREAMBLE
The goals of the Open Font License (OFL) are to stimulate worldwide
development of collaborative font projects, to support the font creation
efforts of academic and linguistic communities, and to provide a free and
open framework in which fonts may be shared and improved in partnership
with others.

The OFL allows the licensed fonts to be used, studied, modified and
redistributed freely as long as they are not sold by themselves. The
fonts, including any derivative works, can be bundled, embedded,
redistributed and/or sold with any software provided that any reserved
names are not used by derivative works. The fonts and derivatives,
however, cannot be released under any other type of license. The
requirement for fonts to remain under this license does not apply
to any document created using the fonts or their derivatives.

DEFINITIONS
"Font Software" refers to the set of files released by the Copyright
Holder(s) under this license and clearly marked as such. This may
include source files, build scripts and documentation.

"Reserved Font Name" refers to any names specified as such after the
copyright statement(s).

"Original Version" refers to the collection of Font Software components as
distributed by the Copyright Holder(s).

"Modified Version" refers to any derivative made by adding to, deleting,
or substituting -- in part or in whole -- any of the components of the
Original Version, by changing formats or by porting the Font Software to a
new environment.

"Author" refers to any designer, engineer, programmer, technical
writer or other person who contributed to the Font Software.

PERMISSION & CONDITIONS
Permission is hereby granted, free of charge, to any person obtaining
a copy of the Font Software, to use, study, copy, merge, embed, modify,
redistribute, and sell modified and unmodified copies of the Font
Software, subject to the following conditions:

1) Neither the Font Software nor any of its individual components,
in Original or Modified Versions, may be sold by itself.

2) Original or Modified Versions of the Font Software may be bundled,
redistributed and/or sold with any software, provided that each copy
contains the above copyright notice and this license. These can be
included either as stand-alone text files, human-readable headers or

in the appropriate machine-readable metadata fields within text or
binary files as long as those fields can be easily viewed by the user.

3) No Modified Version of the Font Software may use the Reserved Font
Name(s) unless explicit written permission is granted by the corresponding
Copyright Holder. This restriction only applies to the primary font name as
presented to the users.

4) The name(s) of the Copyright Holder(s) or the Author(s) of the Font
Software shall not be used to promote, endorse or advertise any
Modified Version, except to acknowledge the contribution(s) of the
Copyright Holder(s) and the Author(s) or with their explicit written
permission.

5) The Font Software, modified or unmodified, in part or in whole,
must be distributed entirely under this license, and must not be
distributed under any other license. The requirement for fonts to
remain under this license does not apply to any document created
using the Font Software.

TERMINATION
This license becomes null and void if any of the above conditions are
not met.

DISCLAIMER
THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND,
EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF
MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT
OF COPYRIGHT, PATENT, TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL THE
COPYRIGHT HOLDER BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY,
INCLUDING ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL
DAMAGES, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING
FROM, OUT OF THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM
OTHER DEALINGS IN THE FONT SOFTWARE.

fonts/provenance.json


```
[
  {
    "file": "literata.woff2",
    "url":
"https://raw.githubusercontent.com/google/fonts/main/ofl/literata/Literata%5Bopsz,wght%5
D.ttf",
    "originalSha256":
"b41138c9373112f32abb589cc22e8674b06ed4048b0c513be922bdd26f274440",
    "conversion": "fontTools TTFont, WOFF2 container conversion; no subsetting"
  },
  {
    "file": "sourcesans3.woff2",
    "url": "https://raw.githubusercontent.com/adobe-fonts/source-
sans/release/WOFF2/VF/SourceSans3VF-Upright.ttf.woff2",
    "originalSha256":
```

"5f16566f7a40d39b339ad26be151fa5a1ab1f0c2574c7a2e619765584a1acbd8",
    "conversion": "None; unmodified upstream WOFF2 release file"
  }
]

fonts/sourcesans3-OFL.txt

Copyright 2010-2020 Adobe (http://www.adobe.com/), with Reserved Font Name 'Source'. All Rights Reserved. Source is a trademark of Adobe in the United States and/or other countries.

This Font Software is licensed under the SIL Open Font License, Version 1.1.

This license is copied below, and is also available with a FAQ at:
http://scripts.sil.org/OFL

SIL OPEN FONT LICENSE Version 1.1 - 26 February 2007

PREAMBLE
The goals of the Open Font License (OFL) are to stimulate worldwide
development of collaborative font projects, to support the font creation
efforts of academic and linguistic communities, and to provide a free and
open framework in which fonts may be shared and improved in partnership
with others.

The OFL allows the licensed fonts to be used, studied, modified and
redistributed freely as long as they are not sold by themselves. The
fonts, including any derivative works, can be bundled, embedded,
redistributed and/or sold with any software provided that any reserved
names are not used by derivative works. The fonts and derivatives,
however, cannot be released under any other type of license. The
requirement for fonts to remain under this license does not apply
to any document created using the fonts or their derivatives.

DEFINITIONS
"Font Software" refers to the set of files released by the Copyright
Holder(s) under this license and clearly marked as such. This may
include source files, build scripts and documentation.

"Reserved Font Name" refers to any names specified as such after the
copyright statement(s).

"Original Version" refers to the collection of Font Software components as
distributed by the Copyright Holder(s).

"Modified Version" refers to any derivative made by adding to, deleting,
or substituting -- in part or in whole -- any of the components of the
Original Version, by changing formats or by porting the Font Software to a
new environment.

"Author" refers to any designer, engineer, programmer, technical writer or other person who contributed to the Font Software.

PERMISSION & CONDITIONS
Permission is hereby granted, free of charge, to any person obtaining a copy of the Font Software, to use, study, copy, merge, embed, modify, redistribute, and sell modified and unmodified copies of the Font Software, subject to the following conditions:

1) Neither the Font Software nor any of its individual components, in Original or Modified Versions, may be sold by itself.

2) Original or Modified Versions of the Font Software may be bundled, redistributed and/or sold with any software, provided that each copy contains the above copyright notice and this license. These can be included either as stand-alone text files, human-readable headers or in the appropriate machine-readable metadata fields within text or binary files as long as those fields can be easily viewed by the user.

3) No Modified Version of the Font Software may use the Reserved Font Name(s) unless explicit written permission is granted by the corresponding Copyright Holder. This restriction only applies to the primary font name as presented to the users.

4) The name(s) of the Copyright Holder(s) or the Author(s) of the Font Software shall not be used to promote, endorse or advertise any Modified Version, except to acknowledge the contribution(s) of the Copyright Holder(s) and the Author(s) or with their explicit written permission.

5) The Font Software, modified or unmodified, in part or in whole, must be distributed entirely under this license, and must not be distributed under any other license. The requirement for fonts to remain under this license does not apply to any document created using the Font Software.

TERMINATION
This license becomes null and void if any of the above conditions are not met.

DISCLAIMER
THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OF COPYRIGHT, PATENT, TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL THE COPYRIGHT HOLDER BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, INCLUDING ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL DAMAGES, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM OTHER DEALINGS IN THE FONT SOFTWARE.

sources/conditionals/NOTICE.txt

```
Sextus Empiricus, Pyrrhoneioi hypotyposeis (Outlines of Pyrrhonism), II.110-114.
Hermann Mutschmann, Sexti Empirici Opera, volume I
(Leipzig: B. G. Teubner, 1912), printed pages 92 and 93.
Public-domain ancient text and printed edition. Mutschmann lived 1882-1918.

Scan: University of Florida, George A. Smathers Libraries, through Internet Archive.
https://archive.org/details/sextiempiriciope01sext
Retained unmodified 2096 x 3177 pixel page-service renditions:
https://archive.org/download/sextiempiriciope01sext/page/n123_w1800.jpg (printed 92)
https://archive.org/download/sextiempiriciope01sext/page/n124_w1800.jpg (printed 93)
Retrieved and inspected 2026-10-31. The printed page numerals, not the filenames
or scan-data page labels, determine these citations: this page service's returned
images were one printed page later than the initial scan-data lookup suggested.
Both complete pages, margins and textual apparatus remain present.

The authority for the excerpts in src/conditional-readings.ts is these images.
They join line endings and normalize punctuation and apostrophes; the fourth
account retains the edition's brackets marking the deletion of axiooma.
Book/section references and physical page numbers are distinct. The English and
teaching Latin are new; Latin abridgments and supplied elliptical words are named
beside their readings. No independent scholarly review is claimed.

The Open Greek and Latin First1KGreek digital text was consulted to locate the
passages; its markup and transcription were not retained as the source witness.
It is attributed to the University of Leipzig and Digital Divide Data, under
CC BY-SA 4.0:
https://github.com/OpenGreekAndLatin/First1KGreek/blob/master/data/tlg0544/tlg001/tlg054
4.tlg001.1st1K-grc1.xml
The displayed Greek was checked directly against the complete printed pages;
printed words missing from digital extraction were restored from those pages.

Sextus reports four competing accounts, rather than endorsing Philo's or
settling the dispute. The connection proponents are unnamed in this passage.
The finite considered-case model, illustration and controls are modern. A
chosen finite table does not certify Diodorus's temporal/modal criterion or
reconstruct the ancient concepts of connection and potential containment.
Code, original English and teaching Latin, diagrams and explanations:
MIT, copyright (c) 2026 Anthony Mercer. Shared font licenses are retained with
Literata and Source Sans 3 in ../../fonts/.
```